



SAGAR CHAUDHARY

Performance Engineering Lead | Site Reliability Engineer | AI/LLM Performance Engineer

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Experience: 5.6 Years • **Open to:** Bangalore, Kolkata, Remote

PROFESSIONAL SUMMARY

Performance Engineering Lead with **5.6 years** of hands-on experience in enterprise performance testing, Site Reliability Engineering (SRE), and AI/LLM observability across insurance, banking, and retail domains. Specialized in designing scalable cloud-native architectures, optimizing end-to-end latency, and ensuring SLA/SLO compliance. Pioneered LLM chatbot performance engineering using LangSmith and WebSocket protocol, tracking TTFT, token throughput, and per-session cost estimation. Expert in **Apache JMeter, LoadRunner, NeoLoad, Dynatrace, Datadog, AWS, Azure, and Kubernetes**. Proven delivery record: **40% database query optimization, 35% JVM heap improvement, 25% login response time reduction, and 71% LLM token cost reduction** across mission-critical production systems.

CORE COMPETENCIES & TECHNICAL SKILLS

Performance Testing	Load Testing, Stress Testing, Spike Testing, Volume Testing, Scalability Testing, Endurance Testing, Soak Testing, Regression Testing, Baseline Testing, Capacity Planning, LLM/Chatbot Performance Testing
SRE & Reliability Engineering	SLI/SLO/SLA Design, Error Budget Management, Incident Management, Root Cause Analysis (RCA), High Availability (HA), Fault Tolerance, Disaster Recovery, Chaos Engineering Principles, Scalability Engineering, Observability, MTTR/MTBF Optimization
AI / LLM Observability	LangSmith, TTFT (Time to First Token), Token Throughput, Input/Output Token Analysis, Prompt Compression, Cost Estimation & Optimization, WebSocket Performance, RAG Performance Testing, GPT-4o / Claude / Gemini Benchmarking
Performance Test Tools	Apache JMeter (Advanced), LoadRunner, NeoLoad, Gatling, OctoPerf, BlazeMeter, K6, Swagger/OpenAPI, Postman, JMeter MCP Server
APM & Monitoring	Dynatrace, Datadog, AppDynamics, New Relic, Grafana, Kibana, Prometheus, ELK Stack, AWS CloudWatch, Azure Application Insights, LangSmith
Cloud Platforms	AWS (EC2, S3, EKS, CloudWatch, Auto Scaling, IAM, Lambda), Microsoft Azure (AKS, Application Insights, Azure DevOps), Kubernetes (EKS/AKS), Docker
CI/CD & DevOps	Jenkins, GitLab CI/CD, GitHub Actions, Azure DevOps, Docker, Kubernetes, Terraform Basics, Shift-Left Performance Testing, Continuous Performance Testing
Scripting & Programming	Java, Python, Groovy, JSR223, Shell Scripting, PowerShell, BeanShell, SQL
Databases	Oracle, SQL Server, MySQL, PostgreSQL, MongoDB – Query Optimization, Index Tuning, Execution Plan Analysis, Lock Contention Analysis
Leadership & Methodologies	Team Leadership, Stakeholder Management, Cross-functional Collaboration, Agile/Scrum, Shift-Left Testing, Mentoring, Technical Presentations

PROFESSIONAL EXPERIENCE

- Senior Associate - Performance Engineering

PwC India • Bangalore, India

March 2026 – Present

 - Lead performance engineering strategy for enterprise client engagements, improving user experience and system reliability across AI-powered and traditional cloud-native applications.
 - Pioneered LLM chatbot performance testing framework using **LangSmith** and **WebSocket** protocol to simulate real-time conversational load and measure AI inference performance under 200+ concurrent users.
 - Defined and tracked key LLM observability metrics including **TTFT (Time to First Token), total token latency, average tokens per request, input/output token counts**, and per-session cost estimation.
 - Reduced P99 latency from **7s to 1.8s** and achieved **71% token cost reduction** through prompt compression (850 → 120 tokens) and max_tokens capping at 250.
 - Configured **dual-platform observability** integrating LangSmith for trace-level LLM analysis and Datadog for infrastructure monitoring, enabling end-to-end performance visibility across the stack.
 - Benchmarked **5 LLM models** (GPT-4o, GPT-4o-mini, Claude 3 Sonnet, Claude 3 Haiku, Gemini 1.5 Flash) for an insurance chatbot, recommending Claude 3 Haiku as the production model based on lowest TTFT (0.7s), cost, and stability.
 - Developed SLI/SLO/SLA monitoring dashboards using Dynatrace and Datadog, ensuring 99.9% availability compliance across production services.
 - Integrated performance testing into Agile CI/CD pipelines using Jenkins and GitHub Actions, enabling shift-left performance engineering.
- Performance Test Engineer

Capgemini • India

2025 – 2026

 - Designed end-to-end performance testing strategy for **RLI Insurance Portal** covering load, stress, spike, and endurance scenarios, ensuring high availability under peak production load.
 - Analyzed APM metrics including CPU utilization, JVM garbage collection, API response latency, and database lock contention using **Dynatrace** and **Datadog**.
 - Reduced login response time by **25%** through targeted JVM tuning, thread pool optimization, and microservices-level refactoring, directly improving end-user experience.
 - Leveraged AI-assisted tooling with Swagger/OpenAPI specifications to accelerate JMeter script generation and critical user journey identification, cutting script development time by **40%**.
 - Validated AI-generated test artifacts manually to ensure accuracy and compliance with performance KPIs and business SLAs.

- **Executed** high-volume performance tests for a US Banking client, simulating **50,000+ concurrent users** using JMeter and LoadRunner on distributed architecture.
- **Implemented** distributed JMeter execution on **AWS EC2** with Auto Scaling for elastic load generation, reducing test infrastructure costs by **30%**.
- **Architected** microservice latency and infrastructure health dashboards integrating AWS CloudWatch, Datadog, and AppDynamics for real-time incident detection and RCA.
- **Delivered 40% improvement** in database query performance through SQL optimization, composite index tuning, and execution plan analysis on Oracle.
- **Led** regulatory performance testing initiatives ensuring SLA compliance and supporting audit-ready documentation for banking performance thresholds.
- **Mentored** 4 junior engineers on JMeter scripting, correlation, and APM analysis best practices.

- **Performed** scalability and load testing for **OmniStore retail microservices** (15+ services) on Kubernetes EKS and AKS using JMeter and Dynatrace, ensuring reliability under peak traffic.
- **Identified and resolved** GC pauses, memory leaks, and thread contention through deep JVM profiling, reducing production incidents by **45%**.
- **Achieved 35% JVM heap optimization** and validated Kubernetes Horizontal Pod Autoscaler (HPA) behavior under sustained and spike load conditions.
- **Built** automated performance regression test suites integrated with **Jenkins** and **GitLab CI/CD** pipelines to enable shift-left performance engineering.
- **Configured** Kafka consumer lag monitoring and Grafana dashboards with Dynatrace for real-time observability and proactive incident detection.

KEY PROJECTS

LLM Chatbot Performance Engineering – PwC AI Observability Initiative

Tech Stack: JMeter • WebSocket • LangSmith • Datadog • Dynatrace • Python • Azure AKS • Groovy JSR223

- Designed performance testing framework for enterprise LLM chatbot using WebSocket protocol to simulate real-time conversational load at scale.
- Tracked TTFT, total response latency, tokens per request, input/output tokens, and per-session cost via LangSmith and custom Datadog histograms.
- **Outcome:** Improved TTFT from 7s to 1.8s (**74% reduction**) and cut per-session LLM cost by **71%** via prompt compression.

RLI Insurance Portal – End-to-End Performance Engineering

Tech Stack: JMeter • OctoPerf • Dynatrace • Datadog • Azure AKS • SQL Server

- Designed testing strategy covering load, stress, spike, and endurance scenarios with full CI/CD integration via Jenkins.
- Identified bottlenecks across API gateway, microservices, and database tiers under 5,000 concurrent user load.
- **Outcome:** Achieved **25% login response time improvement** through JVM and microservice-level tuning.

US Banking – High-Volume & Regulatory Performance Testing

Tech Stack: JMeter • LoadRunner • AWS CloudWatch • Datadog • AppDynamics • Oracle

- Simulated 50,000+ concurrent users for core banking workflows under peak load on distributed architecture.
- Implemented distributed load generation on AWS EC2 with auto-scaling for elastic, cost-efficient test execution.
- **Outcome:** Delivered **40% database performance improvement** through SQL query optimization and index restructuring.

OmniStore Retail Microservices – Kubernetes Scalability

Tech Stack: JMeter • Dynatrace • Datadog • Kubernetes EKS/AKS • Kafka

- Validated HPA behavior under sustained and spike load for 15+ microservices on Kubernetes.
- Identified critical JVM issues (GC pauses, memory leaks, thread contention) via Dynatrace deep-dive profiling.
- **Outcome:** Achieved **35% JVM optimization** and established performance baselines for regression benchmarking.

RELIABILITY & OBSERVABILITY HIGHLIGHTS

- **SLI/SLO/SLA Design & Monitoring** – defined error budgets and performance thresholds across cloud-native services.
- **Incident Management & RCA** – led post-incident reviews to eliminate recurring performance degradations and reduce MTTR.
- **Distributed Systems Observability** – integrated multi-tool monitoring stacks (Dynatrace, Datadog, LangSmith, Grafana, Prometheus) for unified visibility.
- **High Availability & Resilience** – validated failover, autoscaling, and chaos scenarios under spike and endurance conditions.

EDUCATION & CERTIFICATIONS

B.Tech – Electronics & Communication Engineering
Graduated: 2020

Certifications

- ISTQB Certified Tester Foundation Level (CTFL)
- Microsoft Azure Fundamentals – AZ-900
- Apache JMeter Performance Testing Certification
- Deloitte & Accenture Virtual Job Simulations